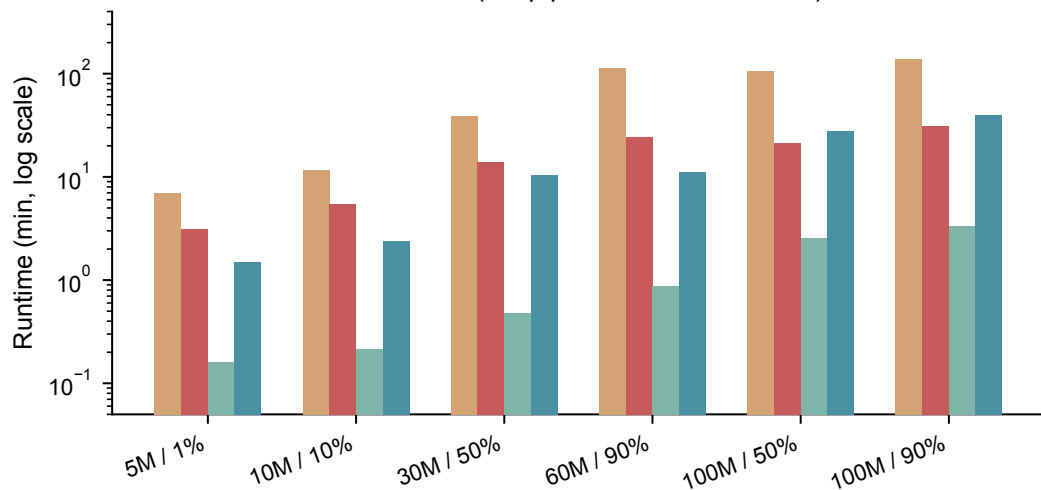
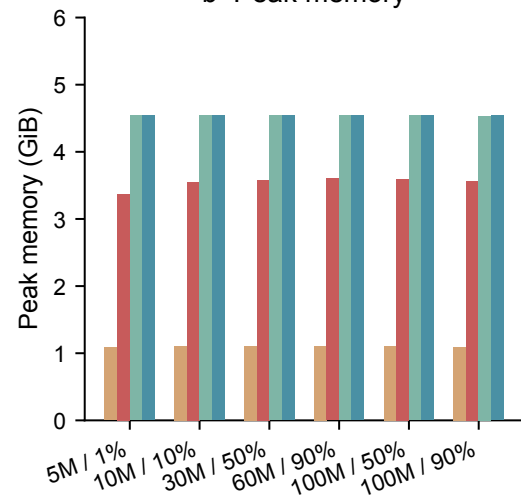


■ KneadData
 ■ Hostile + fastp
 ■ Deacon (depletion only)
 ■ RustyClean AUTO
 (fastp + deacon + conditional Bowtie2 verification)

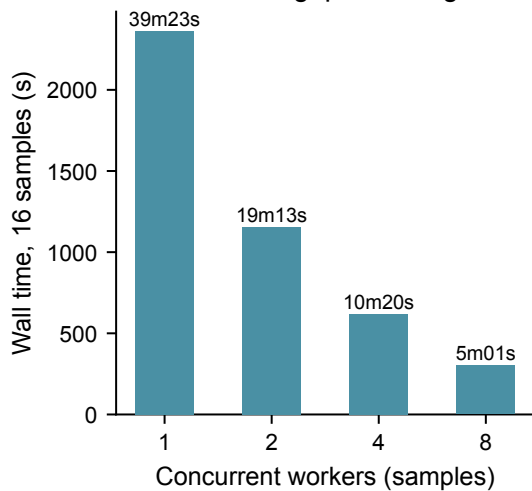
a Runtime (full pipeline or stated basis)



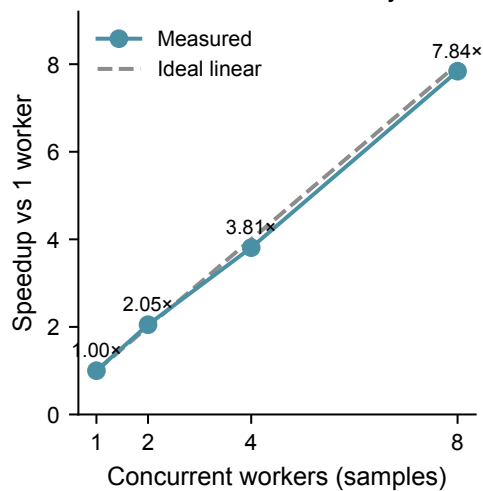
b Peak memory



c Throughput scaling



d Parallel efficiency



e Memory per worker stays flat

